CSCI 611 – Assignment #3
Small Object Detection Using YOLO

1. Dataset Description and Preprocessing

The project used a subset of the Mapillary Traffic Sign Dataset consisting of 18 street-scene images spanning six global regions (Africa, Asia, Europe, North America, Oceania, South America). The dataset covers 37 distinct traffic sign classes including regulatory signs (speed limits, no entry, stop), warning signs (roadworks, slippery surface), and informational signs (parking, pedestrian crossing).

Preprocessing Steps

- Format conversion: Mapillary JSON bounding boxes converted to YOLO format (normalized x_center, y_center, width, height).
- Train/Validation split: 15 training images (83%) / 3 validation images (17%), random seed = 42.
- Directory structure created: images/train, images/val, labels/train, labels/val, and data.yaml config file.
- Quality check: Annotated sample images generated to visually verify bounding box accuracy.

Figure 1: Annotated dataset sample showing ground truth bounding boxes

2. Challenges in Detecting Small Objects

- Limited dataset size (18 images): High risk of overfitting; insufficient diversity in lighting, weather, and viewing angles.
- Class imbalance: Several sign categories had very few examples, making it difficult for the model to learn those classes reliably.
- Scale variation: Signs far from the camera occupy very few pixels, reducing the quality of extracted features.
- Transfer learning gap: The base COCO-pretrained model was optimized for large objects (vehicles, people); traffic signs differ significantly in size and aspect ratio.
- No GPU available: CPU-only training extended experiment turnaround time and limited batch size to 16.

3. Experimental Results

YOLOv8 Nano (yolov8n.pt) was fine-tuned for 30 epochs (early stopping, patience = 10) at 640×640 resolution. Two confidence thresholds were evaluated.

Setting	Model	Detections	Observation
conf=0.5, 640px	Pre-trained (COCO)	0–5 (vehicles, people)	No traffic signs detected
conf=0.5, 640px	Fine-tuned	Traffic sign classes	Specialized for traffic signs, no general objects
conf=0.3, 640px	Fine-tuned	More detections	Lower threshold increases sensitivity

Table 1: Detection results under different settings

Figure 2: Pre-trained YOLO detections (COCO classes, no traffic signs identified)

Figure 3: Fine-tuned YOLO detections (conf=0.5) showing correct traffic sign classes

Figure 4: Fine-tuned YOLO detections (conf=0.3) showing increased recall

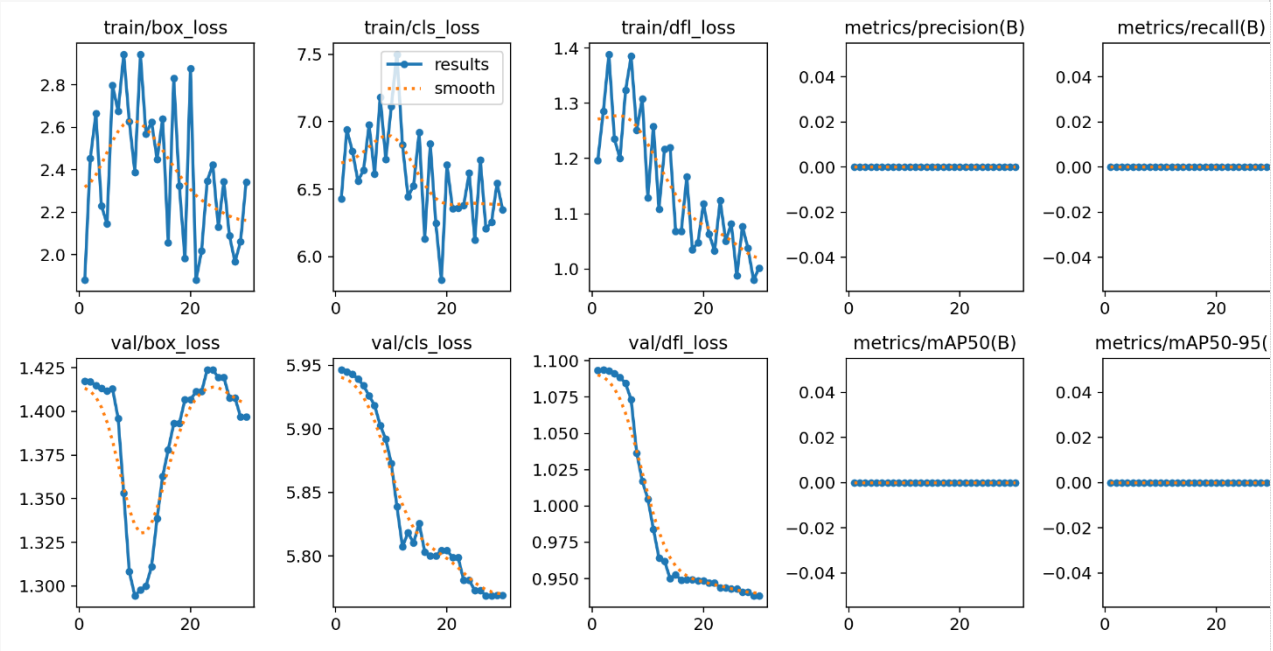

Figure 5: Training loss and precision-recall curves

4. Suggestions for Improving Small Object Detection

- Larger dataset: Expand to 500–2000 images with varied lighting, weather, and camera angles to reduce overfitting and improve generalization.
- Higher input resolution: Training at 1280×1280 preserves more spatial detail for small signs at the cost of additional compute.
- Larger model variant: Upgrading from YOLOv8n to YOLOv8m/l increases feature capacity for fine-grained classification.
- Small-object augmentations: Apply targeted zoom-in crops on sign regions and mix-up/cut-mix techniques to expose the model to more sign instances.
- NMS tuning: Lower the IoU threshold for NMS and explore Soft-NMS to avoid suppressing nearby small detections.
- GPU training: Enables larger batch sizes, faster iteration, and multi-scale training strategies not feasible on CPU.